

AERO-GUARD

Prototype Development — Tech Stack & Roadmap

A practical guide for building the AERO-GUARD compliance system prototype, written for a junior developer joining a tech company. This document covers what AERO-GUARD is, what tools to use, the week-by-week build plan, and the project structure.

Document type	Technical roadmap
Target audience	Junior / entry-level developer
Prototype scope	Provider Hub + Consultant Spoke (mock data)
Timeline	6 weeks

1. What is AERO-GUARD?

AERO-GUARD is a compliance monitoring tool designed for the travel agency industry. It sits on top of the GDS (Global Distribution System) — the terminal-based system that travel consultants use to book flights — and acts as a real-time safety net.

The core problem

When travel consultants make booking errors (missed deadlines, ignored queue messages, duplicate PNRs), airlines charge penalty fees called ADMs (Agency Debit Memos). These fees can cost agencies thousands of dollars per month.

The 5 main objectives

Objective	What it means in simple terms
1. Catch mistakes before they cost money	Real-time pop-up alerts warn consultants the moment they are about to trigger an ADM penalty.
2. Track accountability	Managers see exactly which consultant made what mistake, when it happened, and how much it cost the agency.
3. Help consultants fix problems fast	Built-in helpdesk so consultants get support without leaving their GDS screen. Includes video guides.
4. Speed up data entry	Paste a passport photo and the system reads it (OCR) and auto-fills traveller details into the booking.
5. Handle damage control	When flights are cancelled or changed, managers can quickly issue compensation vouchers to affected clients.

The hub-and-spoke model

The system has two separate interfaces for two different types of users:

Provider Hub (admin side)	Consultant Spoke (user side)
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Used by: System admins, managers, helpdesk staff	Used by: Travel consultants doing daily bookings
Manages agencies, monitors violations across all PCCs, runs financial audits, issues vouchers, provides support, and can remotely emulate any agency PCC.	Receives real-time violation alerts, views personal ADM stats, chats with helpdesk, and uses passport OCR to auto-fill passenger data into GDS PNRs.

2. Tech stack

These tools were chosen because they are beginner-friendly, well-documented, and have large online communities. When you get stuck, you can find answers quickly.

Frontend (what users see)

Tool	Purpose	Why this one?
React	UI framework	Most popular, huge ecosystem, easy to find help online
Tailwind CSS	Styling	Fast styling with utility classes, no custom CSS from scratch
React Router	Navigation	Handles switching between Provider Hub and Consultant Spoke
Lucide React	Icons	Clean, consistent icon set that integrates with React

Backend (the engine)

Tool	Purpose	Why this one?
Node.js + Express	Server	JavaScript on both front and back end - one language to learn
SQLite	Database	No server setup needed. Just a file. Perfect for prototypes.
Socket.io	Real-time alerts	Pushes violation alerts to consultants instantly via websockets

Specialist tools

Tool	Purpose	Why this one?
Tesseract.js	Passport OCR	Free, runs in the browser, reads text from passport photos
Mock JSON data	Fake GDS data	No real GDS connection needed - realistic fake bookings and ADMs

Key principle: No real GDS integration is needed for the prototype. Everything uses mock data that looks and feels like real airline systems. The goal is to demonstrate the concept, not connect to live systems.

3. Development roadmap (6 weeks)

Each week has a clear focus. The order is deliberate — build the foundation first, then add features layer by layer. Do not skip ahead.

Week 1 Foundation

Focus: Set up the project and build the app skeleton

- Initialize the React project with Tailwind CSS
- Set up the Express backend with SQLite
- Build the top navigation bar (Provider / Consultant toggle)
- Build the Provider Hub sidebar navigation
- Create mock data files (agencies, consultants, PNRs, violations, ADMs)
- Create reusable components: MetricCard, Badge, AlertItem, DataTable

Deliverable: Working app shell with navigation and mock data loading

Week 2 Provider Hub - Core

Focus: Build the main admin dashboard pages

- Overview dashboard with metric cards (agencies, violations, ADMs, savings)
- Recent violations feed with severity badges
- Top agencies by ADM exposure ranking
- Agency provisioning page with add/edit/status controls
- Violations page with full log, severity and type filters

Deliverable: Complete Provider overview with agency management and violation monitoring

Week 3 Provider Hub - Support tools

Focus: Build the operational tools managers need

- Audit page with ADM logs and daily/weekly/monthly toggle
- Voucher management (create, send, track client vouchers)
- Support page with open enquiry list and reply functionality
- Broadcast alert tool (send system messages to all agencies)
- PCC Emulation terminal simulating remote GDS access

Deliverable: Fully functional Provider Hub with all 7 sections operational

Week 4

Consultant Spoke

Focus: Build the travel consultant's real-time interface

- GDS terminal mockup (green-on-black with Amadeus-style commands)
- AERO-GUARD extension bar with Alerts / Stats / Helpdesk / Passport tabs
- Real-time violation alerts panel with severity indicators
- Personal stats dashboard (ADMs, revenue saved, weekly trend)
- Helpdesk chat interface with message thread and send button

Deliverable: Working Consultant Spoke showing all 4 panels with mock data

Week 5

The impressive features

Focus: Add the features that make the demo memorable

- Passport OCR tool (drag/paste photo, extract fields, show preview)
- Apply-to-PNR button mapping extracted data to GDS format
- Violation pop-up alert system with timed modal notifications
- Socket.io: Provider actions trigger real-time Consultant updates
- System alert banner for broadcast messages on Consultant side

Deliverable: Full end-to-end flow: violation triggers on Provider, alert on Consultant

Week 6

Polish and demo prep

Focus: Make it presentation-ready

- Visual consistency pass (spacing, colours, typography, icons)
- Add realistic demo data that tells a clear story
- Test every click path (buttons, filters, toggles, navigation)
- Build a 5-minute demo walkthrough script for both interfaces
- Record a screen recording as backup for the live demo

Deliverable: Polished, demo-ready prototype with a rehearsed presentation

4. Project folder structure

This is how the code should be organised. Each component gets its own file. Shared components are reused across both interfaces.

```
aero-guard/  
client/ React frontend  
src/  
  components/  
  Provider/  
    Overview.jsx # Dashboard with metrics  
    Agencies.jsx # Agency provisioning table  
    Violations.jsx # Real-time violation log  
    Audits.jsx # ADM financial reports  
    Support.jsx # Helpdesk + broadcast tool  
    Vouchers.jsx # Voucher management  
    PCCEmulator.jsx # Remote terminal access  
  Consultant/  
    Terminal.jsx # GDS terminal mockup  
    Alerts.jsx # Violation alerts panel  
    MyStats.jsx # Personal ADM dashboard  
    Helpdesk.jsx # Chat with support  
    PassportOCR.jsx # Passport to PNR tool  
  Shared/  
    Sidebar.jsx # Navigation sidebar  
    Topbar.jsx # Top bar with view toggle  
    MetricCard.jsx # Reusable stat card  
    Badge.jsx # Status badge component  
    DataTable.jsx # Reusable data table  
  data/  
    mockData.js # All fake GDS data  
    App.jsx # Main app entry point  
  server/ Express backend  
  routes/  
    agencies.js # Agency CRUD endpoints  
    violations.js # Violation data endpoints  
    vouchers.js # Voucher endpoints  
  data/  
    seed.js # Database seeding script  
    index.js # Server entry point  
    README.md # Project documentation
```

5. Practical tips for a new developer

- **Start frontend-only.** Skip the backend entirely at first. Use mock JSON data directly in React. Add Express later once the UI works.

- **Do not connect to a real GDS.** The prototype only needs to look real. Fake data in JSON files is all you need for a convincing demo.
- **Week 1-2 will take longer than expected.** That is normal. Setting up a project for the first time is always slow. Do not panic.
- **Focus on looking real, not being real.** A client demo cares about how it looks and feels, not how the code is structured internally.
- **Copy the interactive prototype.** The HTML prototype provided with this document can be used as a visual reference or broken into React components directly.
- **Test on the actual demo screen.** Before the demo, run through every click path on the screen you will present on. Layouts can look different on projectors.

End of document